Performance Task 2 – Implementing k-Nearest Neighbors-based Prediction Models

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***Abstract*— This study shows the application of the k-Nearest Neighbors algorithm to classify iris flower species based on the Iris dataset. Based on analyzation, it is shown that the n_neighbors parameter has a significant impact on accuracy, between overfitting and underfitting. The performance of the model is further validated by the metrics and visualizations. Real-world applications of kNN include geotechnical engineering, traffic safety, and MSME financial decision-making to illustrate the adaptability and robustness of the algorithm. Findings confirm kNN's application in classification tasks, but also suggest that data quality and parameter tuning are essential to improve performance and flexibility.**

***Keywords*— *k-Nearest Neighbors, machine learning, Iris dataset, classification, real-world applications*.**

I. Introduction

kNN, or k-nearest neighbor algorithm, is a type of machine learning algorithm that utilizes proximity to compare one data point to a set of data it has been trained upon and memorized to make predictions. This form of instance-based learning affords kNN the 'lazy learning' denomination, and allows the algorithm to perform classification or regression problems. kNN functions off of the assumption that similar points can be found near one another. The Nearest-Neighbor Algorithm needs at least these three factors such as a distance metric, a value k indicating the number of neighbors to look at, and the way for aggregating the classes of neighbor points.

The Iris dataset is a classic machine learning and statistics dataset. It was first presented by British statistician and biologist Ronald Fisher in his 1936 paper "The use of multiple measurements in taxonomic problems" and is often used as a beginner's dataset for practicing classification algorithms. The dataset consists of measurements iris flowers from three different species: Setosa, Versicolor, and Virginica. For the dataset, it contains measurements of 150 iris flowers, 50 for each species.

II. Problem Statement

This study aims to aims to study a kNN based prediction model, as well as measuring the researcher's knowledge about the concepts and processes of the kNN algorithm.

III. Code Analyzation

A. What is the purpose of this program? What is it trying to predict?

The program is the machine learning model of classification where the KNN algorithm is employed to predict a species of the iris flower based on its physical aspects. It utilizes the Iris data set, wherein measurements of different species of an iris flower can be found and include Setosa, Versicolor, and Virginica. The dataset contains a total of four numerical features attributed to each and every flower and include sepal length, sepal width, petal length, and petal width. The goal of the program is to train a classifier that can identify the species of an iris flower given its feature values.

Loading in the Iris data and extracting X (feature matrix) and y (target label) sets in preparation for working on the actual task. The program is prepared using an 80-20 division to split out the training versus the test sets. Ensuring that such a model only trained on the majority 80% of available data will give good performance at seeing unseen information 20%. After splitting the data, a K-Nearest Neighbors classifier is created with n_neighbors=5, meaning that the model will classify each test sample based on the majority class among its five closest neighbors in the feature space. The classifier is then trained using the training data.

After training the model, it is used to make predictions on the test data. For assessing performance, the program calculates the accuracy score, which represents the proportion of correctly classified instances. In addition, a classification report is produced that offers detailed metrics of precision, recall, and F1-score for every single one of the three flower species, giving insights into how the model performs distinguishing between different classes. A confusion matrix is also calculated to show the number of correct and incorrect predictions for each class.

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```
Accuracy:  0.9666666666666667
classification report
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        11
           1       1.00      0.92      0.96        13
           2       0.86      1.00      0.92         6

    accuracy                           0.97        30
   macro avg       0.95      0.97      0.96        30
weighted avg       0.97      0.97      0.97        30

Confusion Matrix:
[[11  0  0]
 [ 0 12  1]
 [ 0  0  6]]
```

Fig. 1. Classification Report of the program and the confusion matrix

To improve interpretability, the program included visualizations. A scatter plot is created to show the test samples in a two-dimensional space, where colors represent the predicted class labels. This allows for a visual assessment of how well the classifier has separated the different species. In addition, the confusion matrix is visualized using a heatmap, which makes it easier to identify misclassifications.

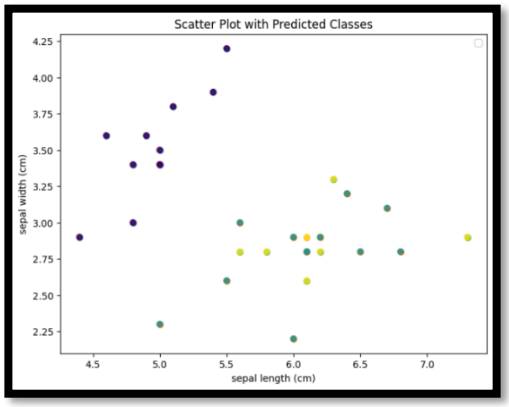

Fig. 2. Scatter Plot with the Predicted Classes

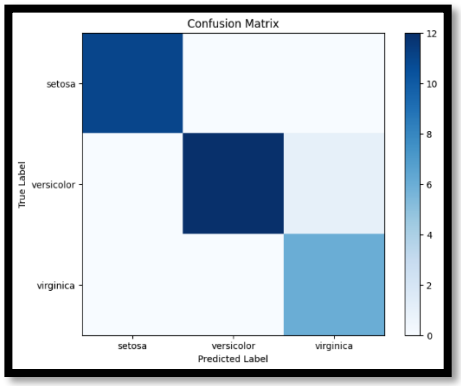

Fig. 3. Visualization of the Confusion Matrix

B. How does the n_neighbors parameter in the KNN model influence its predictions? Experiment with different values and observe the change in accuracy.

In the KNN algorithm, the number of nearest neighbor points that are considered in classifying a new instance will be based on the value in n_neighbors of the K-Nearest Neighbors algorithm. The model will become noisy if the value of n_neighbors is low, for example, 1 or 3, and would have high variance with possible overfitting, fitting the training data really well but does not work well on new data. On the contrary, if it makes a higher value, such as 10 or 15, then the model will be smoother and more general, reducing variance but potentially missing more refined patterns that could lead to underfitting.

The test code below checks the accuracy of the model on classifying iris flowers for various values of n_neighbors ranging from 1 to 100. This is a high variance model (overfitting) when n_neighbors is small and moves to low variance (underfitting) when it is high, with an optimal range in between. Lower values such as 1 or 5 take fewer neighbors; hence, the model is very sensitive to individual data points but very noise-prone. Moderate values, for example, 10 to 33, generally result in better generalization by smoothing decision boundaries. However, too high values, for example, 50, 70, or 100, rely on too many neighbors, which can blur class distinctions and degrade accuracy.

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Load Iris dataset
data = load_iris()
X, y = data.data, data.target

# Split data into training and testing sets
X_Train, X_Test, y_Train, y_Test = train_test_split(X, y,
test_size=0.2, random_state=0)

# Test different values of n_neighbors
neighbor_values = [1, 5, 10, 15, 33, 50, 66, 70, 80, 90, 100]
accuracies = []

for k in neighbor_values:
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_Train, y_Train)
    predictions = knn.predict(X_Test)
    accuracy = accuracy_score(y_Test, predictions)
    accuracies.append(accuracy)
    print(f"n_neighbors = {k}, Accuracy = {accuracy:.4f}")

# Plot the results
plt.figure(figsize=(8,  5))
plt.plot(neighbor_values, accuracies, marker='o',
linestyle='dashed', color='b')
plt.xlabel("Number of Neighbors (k)")
plt.ylabel("Accuracy")
plt.title("Effect of n_neighbors on Model Accuracy")
plt.xticks(neighbor_values)
plt.grid()
plt.show()
```

Fig. 4. Test code used for experimentation

k Value	Accuracy Value
1	1.0000
5	0.9667
10	1.0000
15	1.0000
33	0.9333
50	0.8667
66	0.9000
70	0.8000
80	0.6000
90	0.6000
100	0.5667

Table 1. Table of tested k values and corresponding accuracy values

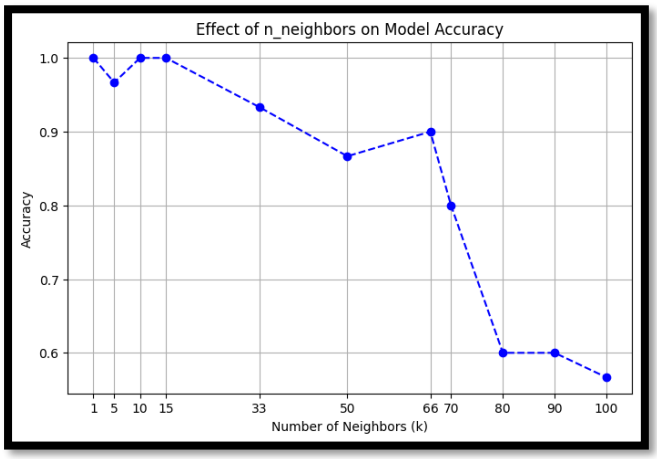

Fig. 5. Visualization of Effect of number of neighbors on Model accuracy

The experiment plots accuracy for different n_neighbors, thereby giving a graphical representation of how the model performs. Most of the time, the best k value will be such that it provides maximum accuracy with a reasonable k value, neither overfitting nor underfitting. Running this test will yield the best k value for the Iris dataset, showing how KNN classification trades model complexity for generalization.

C. . What is the purpose of the confusion matrix? What does it tell you about the types of errors the model makes? Are there any patterns in the misclassified instances?

The confusion matrix of a classification model is determined by comparing actual and predicted labels where errors would reflect. The perfect model produces a diagonal matrix showing the correct prediction wherein all its elements are one, while off-diagonal values show misclassifications, either false positives or false negatives.

The patterns of misclassifications help us look at systematic errors. For instance, if Iris Versicolor and Virginica are confused often by the model, then there is an overlap of feature distributions, which makes it a more challenging task to classify. Such insights will perfect the model through tuning of hyperparameters, like `n_neighbors` in KNN or better selection of features with possibly enhanced feature extraction for much greater accuracy in classification.

IV. Real-world Applications Research

A. Which specific government operations did you investigate in your literature review? What kinds of predictions were being made with KNN in those contexts?

In the development of geotechnical engineering for Valenzuela City, Galupino and Dungca (2022) created a k-Nearest Neighbor model to predict Soil Penetration Test N-values across different Barangays. The researchers used borehole data, wherein latitude, longitude, and depth were inputted, thus resulting in the good prediction of the soil parameters relevant to foundation design and disaster risk reduction. The model had an accuracy rate of 84.4% and provided significant insights for government zoning, land use planning, and minimization of ground-related disaster risks.

Simmachan et al. (2024) developed a k-Nearest Neighbor model using government data from 2019 to 2021 to research the severity of road traffic accidents in Thailand in the time of the COVID-19 pandemic with focus on features, including type of vehicle, region, crash type, weather, and time of incidence using feature selection techniques for further improvement in model performance. The Mean Decrease in Accuracy (MDA) and Mean Decrease in Impurity (MDI) methods were used to identify the most significant features impacting accident severity. The best model of KNN-MDA delivered 81.69% accuracy along with an F1-Score of 89.59% in the pre-pandemic duration, and the best model of KNN-MDI delivered 80.71% accuracy along with an F1-Score of 89.19% during the pandemic. In these findings, it is established that vehicle type constitutes an important element influencing accident severity. The finding would be significant for the proper formulation of road safety management policies and improved public safety measures.

Finally, a literature review by Nawawi and Suhendar (2024) focused on government operations supporting Micro, Small, and Medium Enterprises (MSMEs) especially in terms of finance assistance as well as loan provision. Based on the data given, K-Nearest Neighbors (KNN) provided excellent results in classifying whether the concerned MSMEs are loanable or not. The respective model attained an accuracy rate of 83.939%. It categorized 929 MSMEs as loanable while assigning 170 as the non-loanable MSMEs, streamlining the loan sanctioning process.

Another study found the efficacy of KNN in classification for aid receivers with a highly impressive accuracy level of 98.46%. This outcome depicts KNN as an improvement potential for the decision-making on disbursal of financial aid. It also made use of KNN for predictions of future sales targets of MSMEs that could be benefited by the formulation of business strategies. The outcome signifies a crucial contribution from KNN to

bettering the efficiency of schemes put in place by the government toward MSME development and ease in access to financial services.

B. Based on your review, how did the performance of KNN models vary across different Government operations? What factors seemed to influence their effectiveness?

1. Development of a k-Nearest Neighbor (kNN) Machine Learning Model to Estimate the SPT N-Values of Valenzuela City, Philippines

In their paper to estimate SPT N-values in Valenzuela City, Galupino and Dungca (2022) achieved 84.4% accuracy using the k-Nearest Neighbor model optimized by a hyperparameter k of 5. The high accuracy would mean that the model was efficient in capturing the patterns in the borehole data, which had critical parameters like latitude, longitude, and depth. The generation of 1,912 data layers improved the predictability of the model because it had enabled a better examination of the subsurface conditions within different Barangays.

A significant aspect of the study was the analysis of elevation data, which revealed that the highest elevation point in Valenzuela City is 38 meters above mean sea level, while the average elevation is 2 meters. The kNN model gave the highest elevation at 34 meters with an average of 2.4 meters, these being close to the measured values where the percentage differences are 10% and 16%, respectively. This close alignment points to the accuracy of the modeled topographical features in the region-an important requirement for geotechnical evaluation and infrastructure planning.

Parameters	Measured (m)	k-NN (m)	% Diff.
Maximum Elevation	38	34	10%
Average Elevation	2.0	2.4	16%

Table 2. Comparison of Old and k-NN Generated Parameters

The results reiterate the importance of data quality, because richer and more detailed borehole information significantly contributed to the success in model training. Variability in soil composition from one region to another is also reflected in the model output, an indication of adaptability to unique geological contexts. Utilizing the split dataset approach, the validation procedure proved the adequacy of the model in generalization and robustness to unseen data; this model would be accepted as a tool for making reasonably reliable geotechnical assessments. Overall, the key findings indicate the potential value of kNN models in giving insights for infrastructure development and disaster risk reduction in complex regions.

2. Impact of COVID-19 Pandemic on Road Traffic Accident Severity in Thailand: An Application of K-Nearest Neighbor Algorithm with Feature Selection Techniques

Simmachan et al. (2024) attained an accuracy of 81.69% by means of the KNN model optimized using the MDA pre-pandemic period method within their study on predicting the severity of road traffic accidents in Thailand. This high accuracy reflects that the model has captured the patterns in the road traffic accident data, which consisted of critical features such as vehicle type, region, crash type, weather, and time of incidence. The KNN-MDI model also did well during the pandemic, achieving an accuracy of 80.71%, showing that the model adapts to the changing traffic conditions influenced by government restrictions like lockdowns and social distancing.

Model	2019: Pre-Pandemic Period				2020-2021: Pandemic Period			
	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
LR-Full	74.48	75.27	97.11	84.81	63.96	63.95	98.82	77.65
LR-Step	74.04	74.88	97.02	84.52	61.29	61.25	98.72	75.59
RF-Full	76.22	76.96	**97.36**	85.97	63.25	63.32	98.65	77.14
RF-MDA	73.48	74.06	97.26	84.09	64.08	64.21	98.60	77.78
RF-MDI	78.11	79.18	97.16	87.25	64.92	65.08	98.61	78.41
KNN-Full	79.11	80.28	97.16	87.92	80.23	80.81	98.78	88.89
KNN-Step	76.62	77.50	97.24	86.25	78.14	78.73	98.67	87.58
KNN-MDA	**81.69**	**83.25**	96.96	**89.59**	78.14	78.73	98.67	87.58
KNN-MDI	78.35	79.45	97.16	87.41	**80.71**	**81.28**	**98.80**	**89.19**

Table 3. Performance of sub-models, with kNN algorithm outperforming most of the models

The most influential factor affecting accident severity was identified to be vehicle type, followed by region and crash type. The overall dataset was helpful in improving the effectiveness of KNN models since it allowed a more nuanced investigation of how different factors were contributing to accident outcomes during varying operational contexts. Results thus pointed to data quality and feature selection as of particular importance, for richer datasets that include relevant features and improved model performance. The outcomes of the research are indicative of the potential utility of KNN models in arriving at useful inferences for framing sound management policies for road safety, more specifically in the light of the ongoing dynamics of behavior regarding traffic during the pandemic. Overall, the relevant findings suggest the KNN model can be a safe tool for predicting accident severity and promoting public safety measures in complex environments.

3. Implementation Of MSME Credit Loan Determination Using Machine Learning Technology With kNN Algorithm

In the prediction of loan eligibility for MSMEs, the KNN model had an accuracy rate of 83.939%, which showed its strong capability in classifying MSMEs based on their financial data. On the other hand, in the case of classifying aid recipients, KNN showed a much higher accuracy of 98.46%, which indicates that the model was particularly effective in this scenario, possibly due to the nature of the data or the specific criteria used for

classification. In addition, although accuracy metrics for sales target predictions were not provided, the use of KNN in this area shows its versatility in business performance forecasting.

Several factors have influenced the effectiveness of the KNN models. The quality and relevance of input data most notably affected the effectiveness of the model, such as the inclusion of key variables like length of business operation, number of workers, assets, and net turnover considerably impacted the output of the model. Another critical selection was the choice of k value in the kNN algorithm. Different values of k were tried, and from those experiments, it shows that the precision rates mostly improved when the K is increased. In detail, the highest accuracy was achieved with a K value of 1, where it reached 83.939%. The accuracy decreased to 70.303% with a K value of 10, and to only 57.898% with a K value of 100. This trend shows that a lower K value would make the model more sensitive to the training data, thus making it more effective in its predictive ability. Complexity in the classification task also played a role in determining model performance, since accuracy was better for aid recipients, which might have had more well-defined criteria than the much more subtle assessments that needed to be made for loans. Overall, these differences in performance reflect the role of data quality, choice of K, and task complexity in determining the performance of KNN models on different government operations.

C. Imagine you're a government official involved in one of the operations you reviewed. Would you feel comfortable relying on KNN predictions for decision-making? Why or why not?

Being a government official involved in one of the operations reviewed above, my comfort level in relying on predictions from KNN for decision-making would largely depend upon various parameters. For example, in geotechnical engineering, a KNN model correctly predicted up to 84.4% of soil parameters; hence I would be more comfortable recommending such infrastructure with the use of this aspect for disaster risk reduction also. The quality of the input data is critical; if the data is comprehensive and up-to-date, I would be more inclined to trust the model's predictions, such as the 83.939% accuracy achieved in MSME loan eligibility assessments. High performance metrics, like the 98.46% accuracy for predicting aid recipients, would further instill confidence. However, I would be very cautious for more complex decisions that really require a profound understanding, to recommend combining KNN predictions with expert judgment. Further, the model should continuously be monitored and updated to ensure its relevance as well as its accuracy over time. In general, although I would feel comfortable using the KNN predictions, I would still approach them with a more balanced perspective-which was really emphasizing data quality and the complexity of the decisions.

V. Conclusion

This paper demonstrated the applicability of implementing the kNN algorithm in determining the classification of iris flower species using the Iris dataset. Given the value of n_neighbors and careful consideration on the choice, the study emphasized how model complexity often relates to the generalization as well as which k-value was optimal for both overfitting and underfitting.

The study also displayed several evaluation metrics, such as accuracy scores, confusion matrices, and visualizations, which together validated the performance of the kNN model. The insights obtained reflect the potential of the algorithm to address real-world prediction challenges across domains, as can be seen from its application in geotechnical engineering, traffic accident analysis, and MSME financial assessments. It illustrates how kNN can support practical decision-making, provided that the data quality is robust and the model is carefully tuned.

These results therefore verify that kNN is valid in specific contexts, but they also emphasize the importance of data quality, feature selection, and continuous model optimization. Future work about the algorithm could include the integration of kNN with more sophisticated machine learning techniques that would improve its forecasting accuracy and adaptability in dynamic environments.

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